



Report No: ICR00021

IMPLEMENTATION COMPLETION AND RESULTS REPORT
(Loan No. 5803-JM)

ON A
GRANT

IN THE AMOUNT OF US\$16.365 MILLION

TO
JAMAICA

FOR

Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks

December 16, 2024

Finance, Competitiveness and Innovation
Latin America And Caribbean



The World Bank

Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks (P173012)

ICR DOCUMENT

Regional Vice President:	Carlos Felipe Jaramillo
Country Director:	Lilia Burunciuc
Regional Director:	Oscar Calvo-Gonzalez
Practice Manager:	Yira J. Mascaro
Task Team Leader (s):	Raquel Alejandra Letelier, Eva M. Gutierrez
ICR Main Contributor:	Douglas Randall

**ABBREVIATIONS AND ACRONYMS**

CCRIF	Caribbean Catastrophe Risk Insurance Facility
CCRIF SPC	Caribbean Catastrophe Risk Insurance Facility Segregated Portfolio Company
Cat bond	Catastrophe Bond
CPF	Country Partnership Framework
DPF	Development Policy Financing
DRF	Disaster Risk Financing
ESMF	Environmental and Social Management Framework
E&S	Environmental and Social
FCI	Finance Competitiveness and Innovation
GDP	Gross Domestic Product
GoJ	Government of Jamaica
GRiF	Global Risk Financing Facility
GRM	Grievance Redress Mechanism
IADB	Inter-American Development Bank
IBRD	International Bank for Reconstruction and Development
IMF	International Monetary Fund
IPF	Investment Project Financing
LCR	Latin America and Caribbean
MDTF	Multi-donor Trust Fund
MoFPS	Ministry of Finance and the Public Service
MoF	Minister of Finance
NNDRF	National Natural Disaster Reserve Fund
PDO	Project Development Objective
RTA	Risk Transfer Agreement
SPV	Special Purpose Vehicle
TA	Technical Assistance
TC-W	Tropical Cyclone-Wind
WB	World Bank
WBG	World Bank Group



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**DATA SHEET****BASIC DATA****Product Information**

Operation ID P173012	Operation Name Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks
Product Investment Project Financing (IPF)	Operation Short Name Jamaica Catastrophe Bond
Operation Status Closed	Approval Fiscal Year 2021
Current ESRC Low	

CLIENTS

Borrower/Recipient Jamaica	Implementing Agency Ministry of Finance and the Public Service
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DEVELOPMENT OBJECTIVE

Original Development Objective (Approved as part of Approval Package on 23-Jun-2021)

To expand Jamaica's financial protection against losses arising from severe tropical cyclones-wind.

FINANCING

Financing Source	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Administered Financing	16,365,000.00	16,334,361.11	16,334,361.11



The World Bank

Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks (P173012)

ICR DOCUMENT

TF-B5803	16,365,000.00	16,334,361.11	16,334,361.11
Non-World Bank Financing	150,000.00	0.00	0.00
Borrower/Recipient	150,000.00	0.00	0.00
Total	16,515,000.00	16,334,361.11	16,334,361.11

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Type	Amount Disbursed (US\$M)	Key Revisions
17-Jun-2024	Portal	16.33	• Components • Results

KEY DATES

Key Events	Planned Date	Actual Date
Concept Review	06-Jan-2020	16-Dec-2019
Decision Review	20-Apr-2021	20-Apr-2021
Authorize Negotiations	26-May-2021	27-May-2021
Approval	30-Jun-2021	23-Jun-2021
Signing	25-Jun-2021	25-Jun-2021
Effectiveness	25-Jun-2021	25-Jun-2021
ICR/NCO	20-Dec-2024	--
Restructuring Sequence.01	Not Applicable	17-Jun-2024
Mid-Term Review No. 01	16-Nov-2022	31-Mar-2023
Operation Closing/Cancellation	21-Jun-2024	21-Jun-2024

RATINGS SUMMARY

Outcome	Bank Performance	M&E Quality
Highly Satisfactory	Highly Satisfactory	High

**ISR RATINGS**

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	08-Oct-2021	Highly Satisfactory	Highly Satisfactory	16.33
02	20-Mar-2022	Highly Satisfactory	Highly Satisfactory	16.33
03	23-Sep-2022	Highly Satisfactory	Highly Satisfactory	16.33
04	04-Apr-2023	Highly Satisfactory	Highly Satisfactory	16.33
05	09-Oct-2023	Highly Satisfactory	Highly Satisfactory	16.33
06	30-Apr-2024	Highly Satisfactory	Highly Satisfactory	16.33

SECTORS AND THEMES**Sectors**

Major Sector	Sector	%	Adaptation Co-benefits (%)	Mitigation Co-benefits (%)
FY17 - Financial Sector	FY17 - Insurance and Pension	100	100	0

Themes

Major Theme	Theme (Level 2)	Theme (Level 3)	%
FY17 - Environment and Natural Resource Management	FY17 - Climate change	FY17 - Adaptation	100
FY17 - Finance	FY17 - Finance for Development	FY17 - Disaster Risk Finance	100
FY17 - Urban and Rural Development	FY17 - Disaster Risk Management	FY17 - Disaster Preparedness	100

**ADM STAFF**

Role	At Approval	At ICR
Practice Manager	Yira J. Mascaro	Yira J. Mascaro
Regional Director	Robert R. Taliercio	Oscar Calvo-Gonzalez
Global Director	Jean Pesme	Jean Pesme
Practice Group Vice President	Indermit Gill	Pablo Saavedra
Country Director	Tahseen Sayed Khan	Lilia Burunciuc
Regional Vice President	Carlos Felipe Jaramillo	Carlos Felipe Jaramillo
ADM Responsible Team Leader	Eva M. Gutierrez	Raquel Alejandra Letelier
Co-Team Leader(s)		Eva M. Gutierrez
ICR Main Contributor	Douglas Randall	

I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES**A. CONTEXT AT APPRAISAL**

Jamaica is highly exposed to natural disasters, with increasing vulnerability due to climate change. Jamaica is highly exposed to tropical cyclones, earthquakes, droughts, floods, and landslides.¹ At appraisal, Jamaica was rated as the third most exposed country in the world to multiple hazards, with over 96 percent of the country's gross domestic product (GDP) and population at risk from two or more hazards. Climate change increases Jamaica's vulnerability to natural disasters: models predict Jamaica could be impacted by an increased frequency and/or severity of catastrophic natural disasters because of heightened surface temperatures and sea level rise. The 2020 Atlantic hurricane season was the most active and the fifth costliest season on record and the fifth consecutive above-average season since 2016. Expectations of above-normal activity extended to the 2021 season.²

Increased exposure to natural disasters threatens Jamaica's hard-won macroeconomic stability gains and national development strategy. After decades of chronic macroeconomic imbalances, in 2013 the Government of Jamaica (GOJ) embarked on a program aimed at stabilising the economy, reducing debt, and fuelling growth. The program, underpinned by an IMF Stand-By Arrangement, led to a 50-percentage point decline in public debt accompanied by low and stable

¹ High exposure is attributed to the country's location in the Atlantic Hurricane Belt, the geophysical orientation of its low-lying coastal zones, and its mountainous topography. The Jamaican territory is also crossed by five major fault lines, including the Plantain Garden Fault Zone, which triggered the 2010 Haitian earthquake.

² On December 9, 2020, the Tropical Storm Risk (TSR) issued an extended range forecast for the 2021 season, predicting activity that is above normal. In this report, the organization predicts 16 named storms, 7 hurricanes and 3 major hurricanes. The main factor behind their prediction is the expected development of a weak *La Niña* by the third quarter of 2021. TSR is a venture developed from a UK government-supported project to predict and map tropical storm activity worldwide.



inflation. In 2019, rating agencies Standard and Poor's Global Ratings, Fitch Ratings, and Moody's Corporation upgraded the country's sovereign rating to B+, B+, and B2, respectively. Poverty in Jamaica also declined, reaching its lowest level in a decade in 2018. Yet Jamaica's exposure to natural disasters poses a significant threat to its macroeconomic stability and outlook.³ The World Bank (WB) estimated that Jamaica's contingent liabilities for natural disasters occurring one-in-50 years to one-in-100 years range between 7.34 and 12.0 percent of Jamaica's 2020 GDP. Natural disasters continue to increase Jamaica's sovereign debt level, prompting the need for loans and/or tax increases and/or budget reallocation to finance post-disaster expenditures. Natural disasters also have important welfare impacts: the average damaging hurricane reduces Jamaican per capita consumption by approximately 1.1 percent. More destructive events have impacts multiple times this amount.⁴

The GOJ, with the support of the WB, embarked upon a comprehensive strategy toward developing 360-degree resilience to natural disasters and climate shocks. The reforms cover financial, physical, and social resilience and have been supported by the WB through various instruments, including Development Policy Financing (DPF), Investment Policy Financing (IPF), and technical assistance (TA). The National Public Financial Management Policy Framework and the National Natural Disaster Risk Financing (NNDRF) Policy 2021-2026 underpin the GoJ financial resilience aspects of the 360-resilience approach. The policy framework considers cost-effectiveness, timeliness, and sound administrative arrangements for reducing the fiscal impact of natural disasters by proposing a risk-layering strategy that combines risk retention and risk transfer instruments.

The GoJ requested WB support to diversify its disaster risk financing (DRF) instruments to address remaining financial protection gaps and to assess the appropriateness of a Cat bond. At appraisal, the GoJ had three main ex-ante financial instruments available to finance post-disaster public expenditures: (i) a Contingencies Fund for Natural Disasters; (ii) a Parametric Contingent Credit line from the Inter-American Development Bank (IADB) covering earthquakes and tropical cyclones; and (iii) Caribbean Catastrophe Risk Insurance Facility (CCRIF) Segregated Portfolio Company (SPC) insurance policies. During a preceding ASA (P168156) and project preparation, the WB supported Jamaica in (i) assessing the financial protection gaps corresponding to various natural disaster events (tropical cyclone-wind, earthquakes); and (ii) analyzing the comparative value-for-money of various DRF instruments available to Jamaica, including Cat bonds, expanded CCRIF coverage, and greater allocation to the Contingencies Fund.⁵ Following technical analysis and client dialogue during project preparation, a Cat bond was identified as an effective and innovative risk transfer instrument to address Jamaica's financial protection gap related to severe tropical cyclone events occurring one in 20 to one in 100 years.⁶ The analysis found that a Cat bond would complement Jamaica's existing DRF instruments, consistent with its risk layering strategy. It also found that Jamaica's existing DRF instruments provided adequate financial protection against earthquakes and lower severity / higher frequency excess rain events.

³ International Monetary Fund, "2014 Article IV Consultation Fourth Review Under the Extended Fund Facility and Request for Modification of Performance Criteria," IMF Country Report no. 14/169, June 2014.

⁴ Henry M, Spencer N, Strobl E (2019) The impact of tropical storms on households: evidence from Panel data on consumption. Oxf. Bull Econ Stat 82:1–22. <https://doi.org/10.1111/obes.12328>

⁵ The WB team carried out a Financing Gap and Cost-Benefit Analysis to support the GoJ in making informed decisions about its DRF strategy. The analyses were performed for both *Earthquake* and *Tropical Cyclone – Wind* considering emergency phase losses (i.e. not reconstruction losses).

⁶ The analysis showed that for tropical cyclone (wind intensity events, TC-W) related post-disasters emergency expenditure only, the financing gap ranges from zero, for small and very frequent events, to massive gaps related to very infrequent events (with frequency of less than one in hundred years or less) which should not be financed with ex-ante instruments.

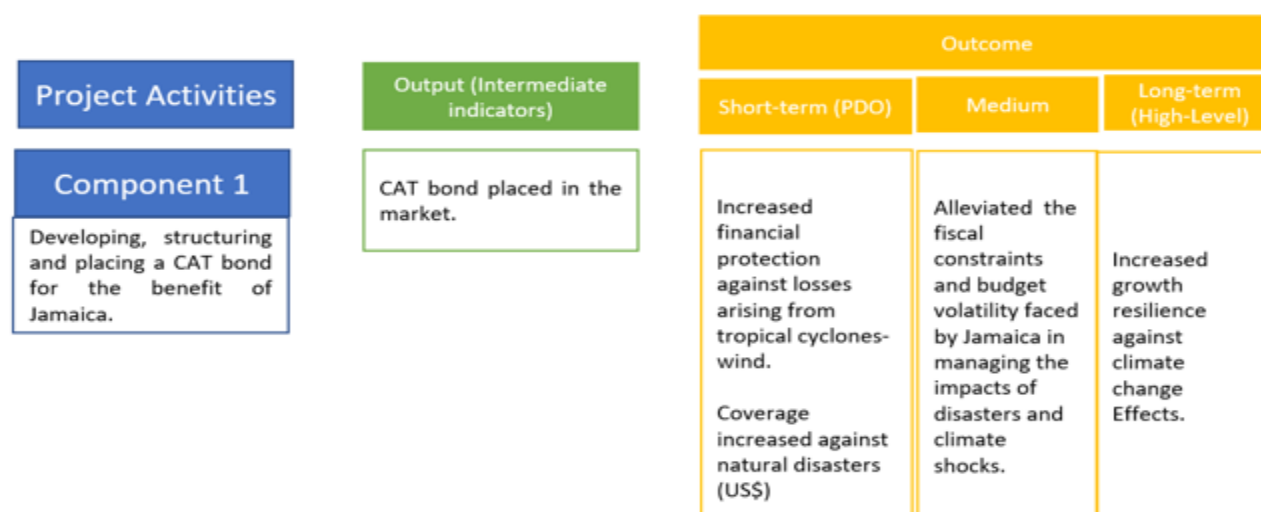


Project Development Objectives (PDO), Theory of Change and Expected Results

The Project Development Objective was to expand Jamaica's financial protection against losses arising from severe tropical cyclones-wind. The long-term outcome of the operation was to increase Jamaica's growth resilience against climate change effects. The Project activities consisted of financing the development, structuring, and placement of a Cat bond (Box 1). The Project's output was the placement of the Cat bond in the market, measured via a binary intermediate indicator (*Cat bond placed in the market: Yes/No*). Conditional on investor interest in the Cat bond, the short-term project outcome (PDO outcome) was to provide increased financial protection against losses arising from tropical cyclones-wind and increase the coverage against those disasters. The achievement of the PDO was measured by the following indicator: *Increased insurance coverage from original transaction, amount USD (target: US\$130 million)*. A second PDO outcome indicator was added during restructuring to reflect the use of project grant balance to contribute to a renewed Cat Bond: *Increased insurance coverage from original transaction, amount USD (target: US\$660,000)*.

As the Cat bond provides coverage for up to three years, over the medium term, Project outcomes also include the alleviation of fiscal constraints and budget volatility arising from natural disasters by providing financial resources to manage emergency expenditures related to those events. The Project's long-term outcome is increased growth resilience as the availability of fiscal resources to deal with disasters is expected to reduced consumption and production losses associated to natural disaster disruptions.

Figure 1: Theory of Change



Components

The project was designed as a recipient executed trust fund (RETF) / IPF with a single component: the development and implementation of risk transfer transactions against tropical cyclones-wind effects. The GoJ decided on its preferred Cat Bond structure following development and analysis of seven prototypes. The Project design benefitted from WB TA and capacity-building on modelling and generation of Cat bond prototypes and advice on best options. The Minister of Finance and MoFPS staff were highly engaged throughout the prototype development and selection process. The tropical cyclone-wind (TC-W) coverage was designed to trigger based on the central pressure of the tropical cyclone as reported



by the U.S. National Hurricane Centre. The proposed solution covered the geographic area of Jamaica using a grid with 21 cells (a.k.a. a “cat-in-the-grid”). If the track of a tropical cyclone crosses through one or more boxes, then a payout might be triggered according to the minimum central pressure of the hurricane in the box. Each box has, at least, two trigger thresholds which trigger different percentage payouts of the Cat Bond based on a stepwise linear function (these thresholds varied across the prototypes).⁷

The Project financed the risk premium and transaction costs of the Cat bond, a risk transfer instrument issued by the WB through its Treasury for the benefit of Jamaica. At appraisal, Jamaica did not have local experience in executing Cat Bond transactions. In order to benefit from the WB experience and expertise in the Cat Bond market as well as its market brand and efficient transaction structure, Jamaica requested that the WB Treasury intermediate the transaction via issuance of a WB Cat bond. Under this arrangement, Jamaica would enter into an agreement with the WB to transfer the catastrophe risk of Jamaica (the “Risk Transfer Agreement”, RTA). Contemporaneously, the WB would intermediate the transfer of Jamaica’s catastrophe risk to capital market investors by issuing a Cat bond.

Box 1: What is a Cat Bond?

A Cat Bond is a financial instrument that allows entities exposed to natural disaster risks to transfer a portion of that risk to bond investors for a period of one to five years. Cat bonds work in a similar manner to insurance in that they pay out when a disaster event meets certain pre-defined criteria or “parameter” (e.g., a specified earthquake magnitude).¹

In a typical Cat bond structure, the entity exposed to the risk (known as the “sponsor”) enters into an insurance contract with a Special Purpose Vehicle (SPV) that issues the Bonds to investors. The SPV invests the proceeds of the bond issuance in highly rated securities that are held in a collateral trust, and it transfers the return on this collateral, together with the insurance premiums received from the sponsor, to the investors as periodic coupons on the Bonds.

If a specified natural disaster occurs during the term of the bond, some or all of the assets held as collateral are liquidated and that money is paid to the sponsor as a pay-out under its insurance contract with the SPV. In such case, the investors lose some or all of the principal (Cat bonds are fully funded transactions). If no specified event occurs, the collateral assets are liquidated on the maturity date of the Bonds and the money is paid to the investors. The global Cat bond market has developed steadily in recent years, with issuances reaching US\$16.4 billion across 95 deals in 2023, with risks concentrated in the United States and dominated by hurricane and earthquake risks.

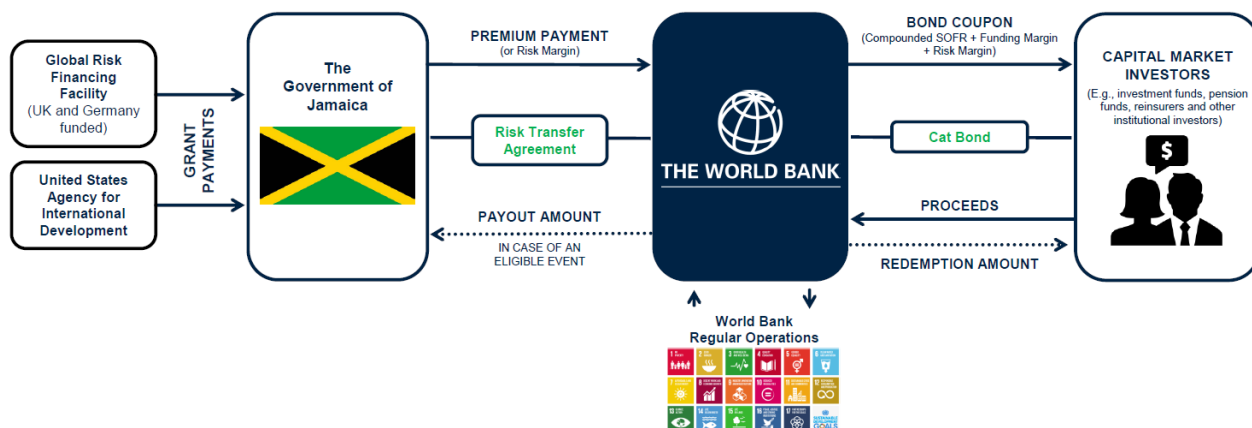
The WB secured donor support to cover the Cat bond risk premium and associated transaction costs. In 2019, the WB began exploring the use of Global Risk Finance Facility (GriF, a multi-donor Trust Fund administered by the WB) funds to pay the risk premiums and transaction costs (US\$16.365 million) of a Cat bond transaction. GriF formally approved the use of resources for this operation in February 2020. The USAID also contributed US\$5 million to the payment of the risk premiums through a parallel project.

⁷ For example, in one of the prototypes, the cell that includes Kingston is parameterized as follows: (i) payout from 30 percent to 70 percent if central pressure is between 969mb and 900mb; (ii) payout from 70 percent to 100 percent payout if central pressure is less than 900mb.



The Project funds were designed to flow from the WB-administered trust funds to the WB Treasury on behalf of the GoJ (the recipient and beneficiary of the coverage) for the purposes of paying the Cat Bond premiums and transaction costs. The Grant under this Project covered the risk premium and transaction costs of the Cat bond. The proceeds of the Grant (US\$16,365,000) would be disbursed from the WB Grant Account to WB Treasury according to the terms and conditions an Engagement Letter signed between the GoJ and the WB Treasury. The Engagement Letter authorized the WB Treasury to prepare the risk transfer transaction and request payment of Project funds to cover risk premium and transaction costs of the Cat bond. The WB Treasury would engage external service providers, such as external legal counsel, lead arrangers (managers, structuring agents, bookrunner agents), event calculation agent, and listing agent to issue the Cat bond and to prepare the legal documentation for the risk transfer transaction. To mitigate any potential or perceived conflict of interest that would have arisen from using Project funds to pay the IBRD intermediation fee, the GoJ allocated budgetary resources to cover this expense (US\$150,000)

Figure 2: World Bank Cat Bond Structure – Jamaica 2021 (original transaction)



Financial resources provided by the Cat bond in the event of a payout were expected to enhance the GoJ's ability to cover post-disaster costs. If a triggering disaster occurs, Jamaica would issue a notice to the independent calculation agent who would determine the payout amount based on the central pressure and track of the cyclone. The bond principal would be reduced by an amount equivalent to the client's insurance payout. The WB would transfer the payouts to the GoJ as soon as a calculation report is available, within weeks of a tropical cyclone event, without the need to assess real losses incurred by the country. The WB would not retain any risk on its balance sheet. Cat bond payouts could be allocated to the general budget or to one of the existing disaster risk funds. As the Project scope is limited to Cat bond design and placement, it does not address the use of Cat bond payouts.

It is important to note that the Project was not designed as a typical IPF. The project represented a novel and innovative use of the IPF instrument to finance and intermediate a risk transaction on behalf of a Recipient country, leveraging donor funds. Indeed, the project would not involve typical IPF implementation (i.e., no activities to be implemented on Jamaican territory or purchasing of goods to be delivered to Jamaica). Rather, implementation of the project would consist of a financial transaction intermediated by the WB Treasury. The WB was authorized to provide these risk intermediation services under its existing policies applicable to disaster risk management products offered to



clients as approved by the Board of Executive Directors, however this represented an expanded role of the WB Treasury in the context of an IPF (see discussion of required waivers in “Issues during Preparation”).

The Project counterpart was the Ministry of Finance and Public Services (MoFPS), albeit with a limited role in implementation. Prior to project approval, the MoFPS would be responsible for requesting the Bank’s intermediation (i.e., making an independent decision on engaging the Bank), providing input to the Bank on the design and objectives of the risk transfer transaction and signing the Risk Transfer Agreements. During project implementation, the MoFPS role will be limited to maintaining monitoring and evaluation arrangements that allow proper reporting and submitting withdrawal requests to the Bank.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION

While the PDO was not revised during implementation, a restructuring of the Project was approved on May 16, 2024 to facilitate a renewal of the Cat bond by the GoJ. The GoJ decided to renew the Cat bond to provide the country with US\$150 million protection against tropical cyclone risk for an additional four consecutive seasons (through end-2027) following the expiration of the original Cat bond. The WB Treasury intermediated the renewal transaction but, unlike the original Cat bond transaction under the Project, the GoJ paid the risk premium and most of the transaction costs with its own funds. The restructuring of the Project enabled the GoJ to use the balance of the Grant proceeds held by the WB (US\$179,102.55) to contribute to costs related to the preparation of the transaction.

Under the restructuring, a minor amendment was made to the Grant Agreement. The new Engagement Letter signed between the GoJ and the WB Treasury on April 2, 2024 to facilitate the Cat bond renewal represented a change in the Implementation Arrangements of the Project, requiring an amendment of the Grant Agreement. Specifically, the definition of “Engagement Letter” in the Appendix of the Grant Agreement was updated to reflect the new Engagement Letter. All other terms and conditions of the Grant Agreement remained in full force and effect. The provisions set forth in the amendment to the Grant Agreement became effective retroactively as of April 2, 2024, upon countersignature by the GoJ on June 13, 2024.

As part of the restructuring, Project Component and Results Framework were also updated to reflect the Cat bond renewal. Counterpart funding was increased by \$40,670,000 as most of premiums and transaction costs of the renewal transaction will be paid by the GoJ, resulting in a proposed cost of \$57,185,000. The amount under Trust Funds was unmodified, as the balance of US\$179,102.55 was used to pay for a portion of the transaction costs. Under the Results Framework, the PDO indicator was split into two sub-indicators, one for the original transaction and another one for the renewal transaction, which was calculated as the proportion of the total increased insurance coverage (US\$150 million) that is attributable to Bank project financing (US\$ 660,000 or 0.44%). The end target-date of the intermediate result indicator (Cat bond placed in the market) was changed to June 2024 to include the renewal transaction.



II. OUTCOME

A. RELEVANCE OF PDO

The relevance of the PDO is rated as *High*.

The Project is strongly aligned with the current Country Partnership Framework (CPF), as well as the CPF in effect at the time of appraisal. The Project objectives support Pillar III of the CPF for FY14-FY17⁸ on Social and Climate Resilience, by expanding the range of financial instruments available to the GoJ to respond to climate change events and natural disasters, thereby building greater financial, social, fiscal, economic growth, and physical resilience. The Project is also well-aligned with Objective 5 (Strengthen Disaster Risk Management) and Objective 6 (Improve Response to Economic Shocks) in the current CPF FY24-FY27. Indeed, the CPF specifically highlights the Cat Bonds supported under the Project as a key instrument in the GoJ's comprehensive disaster risk financing framework.

The Project also contributes to the WB's goals ending extreme poverty on a livable planet and supports the private capital enabling and maximizing finance for development agendas.⁹ Natural disasters disproportionately affect poor and vulnerable populations, and thus the resources enabled under the Project would be essential for recovery efforts to mitigate the impact of tropical cyclones on these groups. Ex-ante financial protection through instruments like the Cat bond are also critical to safeguard the recent macroeconomic and development gains achieved by the GoJ. Further, the Project directly supported the mobilization US\$142.66 million in risk capital from international markets through crowding in capital markets funding to bridge the GoJ's financial protection gap, thereby contributing the WB's private capital enabling agenda. The renewal of the Cat bond by the GoJ, which demonstrated the project's impact on building GoJ's capacity to implement their National Natural Disaster Risk Financing Policy via capital markets also demonstrates the project's alignment with the maximizing finance for development agenda.

The Project is strongly aligned with the Crisis Preparedness and Response (CPR) Toolkit and has indeed informed its evolution and expansion. Cat Bonds are a key tool in the CPR Toolkit. Recent changes reflected in the expanded CRR Toolkit (approved by the Board on February 1, 2024) were informed by the Project, in particular the revision to streamline the use of Bank operations to allow coverage of disaster risk transfer transactions (DRTT) costs via Bank financing proceeds.

B. ACHIEVEMENT OF PDOs (EFFICACY)

The achievement of the PDO (efficacy) is rated as *High*.

The original Cat Bond mobilized US\$185 million from capital markets to provide financial protection for Jamaica against potential economic losses by severe tropical storm winds over three hurricane seasons. The original Cat bond was issued by the WB Treasury in July 2021. The financing and successful placement of the Cat bond achieved the

⁸ The CPF for FY14-FY17 (Report No. 85158-JM) as revised by the CPS Performance and Learning Review (PLR) (Report No. 112663-JM) was extended to 2019 in May 2017.

⁹ Under the then-current PCM guidelines, the Project was not tagged as PCM since it was financed by trust funds and not by the WB's own financing. However, according to the updated Guidance Note "Measuring and reporting World Bank Private Capital Mobilization" from June 8th, 2023, the IBRD/IDA commitment is no longer required to be able to count PCM and PCM from even fully TF-funded projects is possible. Thus if the project was prepared in 2024 it would likely have been tagged as PCM.



PDO of expanding Jamaica's financial protection against losses arising from several tropical storm winds. The decision by the GoJ to renew the Cat Bond with (mostly) its own funds and extend the financial protection for four additional hurricane seasons (providing a further US\$150 million in coverage) affirms the efficacy of the project design and implementation in achieving the PDO. The Cat bond complements Jamaica's portfolio of DRF instruments, providing more complete coverage at higher risk layers and leveraging pre-arranged risk transfer instruments in addition to existing contingent funds and credit lines.

Table 1: Jamaica Original Cat Bond Transaction Summary

Peril	Tropical Cyclone
Principal (Coverage Amount)	US\$185 million
Risk Period	2.4 years (24 July 2021 to 21 December 2023)
Trigger Type	Parametric CAT-in-a-grid
Payout Structure	Linear sliding scale based on severity of storm with a minimum payout of 30% of principal and maximum payout of 100% of principal
Risk Statistics (Annual)	Expected Loss: 1.52%, Attachment Probability 2.37%
Price / Premium Multiple	Risk Margin: 4.4%

Overall, the project directly supported US\$142.66 million in financial protection for Jamaica against potential economic losses by severe tropical storm winds, exceeding its original targets. The primary results indicator (*Increased insurance coverage from original transaction*) reflects the share of the financial protection achieved by the original Cat Bond corresponding to the grant financing provided by GriF under the project (US\$16.365 million, or 77% of the total premium and transaction costs). Following the placement of the original US\$185 million Cat Bond, the indicator reached US\$142 million, exceeding the target of US\$130 million. The remaining share of the financial protection under the original Cat bond is attributed to USAID under a parallel project which provided US\$5 million towards the premium and transaction costs (corresponding to US\$43 million in financial protection achieved). The secondary results indicator refers to the share of the additional financial protection supported by the balance of the Project's grant funds which contributed to the premium and transaction costs of the renewed Cat bond (\$660,000 or 0.44% of the financial protection under the renewed Cat Bond).

Table 2: Results Indicators

Indicator	Baseline (Jun 2021)	Target	Actual at closing (June 2024)
PDO-level			
Increased insurance coverage from original transaction (Amount, USD)	0	130,000,000	142,000,000
Increased insurance coverage from renewal transaction (Amount, USD)	0	660,000	660,000
Intermediate			
Cat bond placed in the market (Yes/No)	No	Yes	Yes



The project also achieved its medium-term outcomes of alleviating fiscal constraints and budget volatility arising from natural disasters. The GoJ has confirmed the efficacy of the Cat bond in contributing to stable fiscal management and reducing budget volatility in the context of increasing exposure to natural disasters. These achievements were also recognized by credit rating agencies: on September 20, 2021, Fitch Ratings issued a “Non-Rating Action Commentary” that the Cat Bond “significantly strengthens the country’s natural disaster risk mitigation strategy.” In a subsequent comment on May 1, 2024, following the issuance of the Cat bond renewal, Fitch Ratings noted that “this financing approach allows Jamaica to transfer some hurricane risk to the bond markets without interrupting its current rapid decline in debt-to-GDP.”

The Project included several innovations which enhanced its efficacy. The original Cat bond was the first Cat bond to use a cat-in-a grid parametric trigger design for tropical cyclone risk. The “cat-in-a-grid” trigger places a grid over Jamaica and surrounding waters with each grid having a centralized air pressure threshold. Payout is triggered if a named hurricane passes through a grid and has centralized air pressure at, or below, the threshold for that grid (the lower the centralized air pressure, the higher the intensity of the hurricane). The size of the payout is related to how many such grids are breached. This transparent and granular approach allows for a tailored approach, for example, providing higher thresholds for geographical areas where losses are likely to be higher from a direct hit, e.g., Kingston and Montego Bay. Relative to other DRF instruments in the GoJ’s disaster risk finance portfolio, the Cat bond’s cat-in-a-grid trigger was simple and transparent to stakeholders (i.e., versus more complex “modeled loss” triggers under CCRIF insurance policies). The Cat bond also included an innovative reporting feature resulting in a quick payout calculation, within weeks of a triggering event.

The project’s achievements have received international recognition among market stakeholders and peer countries. The Jamaica Cat bond won the “Closing the Protection Gap Initiative of the Year” at the 2022 Trading Risk awards ceremony. The Project was featured in a Spring Meetings event in 2022 on regional Cat Bonds in the Caribbean. The Ministers of Finance in Lesotho and Eswatini requested a meeting with the Finance Minister of Jamaica during Spring Meetings 2023 to discuss its experience in disaster risk finance and the role of Cat bonds.

C. EFFICIENCY

The efficiency of the Project is rated as *High*. The Project’s value-for-money proposition was validated by the GoJ’s decision to renew the Cat Bond and to finance the transaction with its own resources (excepting the small share covered by the balance of Grant proceeds held by the WB).

The intermediation role of the WB Treasury in the operation helped to ensure that Jamaica benefitted from competitive pricing in the Cat bond market. The WB has a long track-record of successful transactions in the Cat bond market: as of September 2024, the WB has executed 32 Cat bonds totaling US\$4.7 billion.¹⁰ Investors value that the WB is a known and trusted issuer. Investor demand for WB Cat bonds is also positively impacted by the use of WB-issued Cat bond proceeds to support the financing of sustainable development projects globally (helping them to meet ESG targets). Another key competitive advantage is the risk diversification that the World Bank brings to the market – for example, a Philippines typhoon or a Chilean earthquake have little correlation with the rest of the Cat bond market. As a result of these factors, WB Cat bonds price below the market average for the same level of expected loss, on average.

¹⁰ See the WB Treasury’s Product Note on Disaster Risk Transfer, which also includes a summary of WB intermediated Cat Bonds since 2008: <https://thedocs.worldbank.org/en/doc/46c9b740fe83bc96681ff097681d9e6b-0340012024/original/Disaster-Risk-Transfer-Product-Note.pdf>



The project did not use Jamaica's IBRD lending envelope, thereby reducing the full economic cost of the instrument and helping the GoJ to optimize the use of WB financing. The Cat bond was structured to avoid credit exposure to Jamaica. The Jamaican catastrophe risk was fully passed to investors through the issuance of a WB Cat bond. Thus, there was no use of the WB envelope for Jamaica or counterpart credit exposure. Since project provided fiscal resources to mitigate the impact of natural disasters, it reduces future borrowing needs for this purpose, including from the WB.

Rigorous analytics during project preparation ensured value-for-money and an efficient allocation of resources across a diverse set of complementary DRF instruments to address Jamaica's risk profile. The value-for-money of the project was enhanced by ex-ante analyses undertaken by the WB to (i) assess the financial protection gaps corresponding to various natural disaster events (tropical cyclone-wind, earthquakes); and (ii) analyzing the comparative value-for-money of various disaster risk financing (DRF) instruments available to Jamaica, including Cat bonds, expanded CCRIF coverage, and greater allocation to the Contingencies Fund. The comparative analyses followed best practice methodologies in determining the expected net present value of various DRF instruments for Jamaica's financial protection gaps related to tropical cyclone-wind (no financial protection gap related to earthquakes was identified). The conclusions of these analyses supported the choice of the Cat bond, and the results were widely shared with the client, donors and other project stakeholders, supporting a transparent and evidence-based approach. The GoJ also placed a priority on diversifying its DRF instruments and valued that Cat Bonds offer multi-year financial protection (providing certainty and stability of coverage and pricing).¹¹ The GoJ continued to purchase the same level of coverage under CCRIF, demonstrating that the Cat bond provided complementary financial protection and was not a substitute for other DRF instruments.¹² An ex-post analysis by the WB team concluded that the original Cat Bond (as placed in the market) provided cost-effective financial protection relative to other potential DRF instruments.

Concerns raised by some international stakeholders and the press about Jamaica's Cat bond reflected common misconceptions about the design of parametric risk transfer instruments and their role within broader DRF toolkits.¹³ Some stakeholders and press outlets reacted to the absence of Cat bond pay-outs triggered during hurricanes within the risk periods.¹⁴ The criticisms suggest that the Cat bond triggers are overly strict and that returns to investors are disproportionately high. In the case of Hurricane Beryl, the absence of pay-outs under the Cat bonds were contrasted with the US\$26.6 million pay-out under the GoJ's CCRIF tropical cyclone and excess rainfall insurance policy. However, these concerns misunderstand both the explicit objective and the design of GoJ's Cat bonds: the instruments were designed to provide financial protection for higher risk layer events, i.e. occurring one in 20 to one in 100 years, as a complement to other instruments in the GoJ's DRF toolkit (e.g., CCRIF policies). Return rates on Cat Bonds reflect investment risk (indeed investors can lose the entirety of their investment) and actual pay-outs over the WB's historical portfolio of Cat Bonds suggests good value-for-money for the GoJ: a recent analysis of Cat Bond issuances and payouts between 2017 and September 2024 shows that World Bank Cat bonds have a premium to pay-out ratio of approximately 1.04, that is, USD1 of payouts have been made for every USD1.04 spent on Cat Bond premiums.¹⁵

¹¹ In addition, there was the question about whether CCRIF would be able to offer cost-competitive insurance instruments to address higher-risk layer events (as covered by the Project) or whether CCRIF would be willing to exceed its US\$100 million coverage limit (sum insured) per natural hazard per country (CCRIF has not exceeded this threshold for any country to date).

¹² On June 1, 2021, the CCRIF insurance coverage was kept at about previous year level: US\$ 238.03 million. The coverage was split as follows: Tropical Cyclone=US\$ 81.63 million, Earthquake=US\$ 125 million, Excess Rain=US\$ 31.4 million.

¹³ See, for example, the July 2024 publication by the Climate Vulnerable Forum and Vulnerable Twenty Group, "World Bank Should Course-Correct for More Flexible Cat Bond Trigger Conditions in the Wake of Jamaica's Experience with Hurricane Beryl," and also an article published in Bloomberg on August 18, 2024, "The Huge Profits Investors Have Made on Catastrophe Bonds are Raising Eyebrows."

¹⁴ Including Hurricane Grace in August 2021, Hurricane Ian in September 2022, or Hurricane Beryl in July 2024.

¹⁵ *Demystifying Pre-Arranged Financing for Governments: A Stocktake of Financial Instruments from International Financial Institutions* (2024), Centre for Financial Protection: <https://www.disasterprotection.org/publications-centre/demystifying-pre-arranged-financing-for-governments>



D. JUSTIFICATION OF OVERALL OUTCOME RATING

The overall outcome rating is *Highly Satisfactory*. The overall outcome rating is justified by the following:

- The PDO remains highlight relevant to Jamaica’s development strategy as demonstrated by the Project’s alignment with Objective 5 (Strengthen Disaster Risk Management) and Objective 6 (Improve Response to Economic Shocks) in the current CPF FY24-FY27.
- Cat bonds are cited as a key DRF instrument in Jamaica’s National Natural Disaster Risk Financing Policy, complementing other instruments including those from the WB (i.e. DPF Cat DDO under preparation at time of ICR) and CCRIF insurance policies.
- The Project is strongly aligned with WBG corporate strategy and priorities, including the private capital enabling agenda as well as the Crisis Preparedness and Response Toolkit. Indeed, the project informed the 2024 expansion of the Toolkit which streamlined the preparation and design of disaster risk transfer transactions as part of lending operations.
- The PDO outcome was achieved via the successful placement of the original Cat Bond in international markets, directly contributing to the mobilization of US\$142 million from private investors for financial protection against potential economic losses by severe tropical storm winds and exceeding the PDO target.
- The achievement of the project’s medium-term outcomes of alleviating fiscal constraints and budget volatility arising from natural disasters was affirmed by credit rating agencies.
- The renewal of the Cat Bond by the GoJ via (almost entirely) their own funds, validates the relevance, efficacy, and efficiency of the Project. The Cat Bond renewal also demonstrates that the Project provided institutional strengthening to the MoFPS by enhancing their technical capacity and financial willingness to expand their participation in the Cat Bond market.
- The value-for-money of the project design was enhanced by technical support and capacity-building in a preceding TA engagement and during project preparation, in particular, WB support to the GoJ in assessing the financial impact of natural disaster events (tropical cyclone-wind, earthquakes) and in analyzing the comparative value-for-money of various disaster risk financing (DRF) instruments available to Jamaica, including Cat bonds, expanded CCRIF coverage, and greater allocation to the Contingencies Fund. The conclusions of these analyses supported the choice of the Cat bond, and the results were widely shared with the client, donors and other project stakeholders, supporting a transparent and evidence-based approach.
- The intermediation role of the WB Treasury helped to ensure that Jamaica benefitted from competitive pricing in the Cat bond market and the project structure reduced the full economic cost of the instrument by not using Jamaica’s IBRD lending envelope.
- The operation included several innovations (e.g., first use of a Cat-in-the-grid trigger structure for a Cat bond); the original Cat bond was the first Cat bond independently sponsored by a Caribbean or small-island country.



III. KEY FACTORS AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

The innovative nature of the project and the market impacts of the Covid-19 pandemic resulted in an extended preparation period. The Project Concept Note (PCN) review meeting took place in December 2019. Project preparation was subsequently halted due to COVID-related disruptions in the insurance linked securities market in the spring of 2020. While the WB has issued many Cat bonds for client countries, this was the first Cat bond intermediated and financed via an IPF instrument underpinned by donor grants funds (i.e. a recipient executed trust fund (RETF) / IPF). Thus, several waivers were required to adapt existing IPF guidelines to the project. Project preparation resumed in early 2021, with a Decision Review on April 20, 2021 and Board Approval on June 23, 2021.

Waivers were obtained to reflect the expanded role of WB Treasury and the limited role of MoFPS in project implementation. The project design featured a novel application of an IPF instrument to finance and intermediate a risk transaction. The project would not involve typical IPF implementation (i.e., no activities to be implemented on Jamaican territory or purchasing of goods to be delivered to Jamaica), but rather project implementation would consist of a financial transaction intermediated by the WB Treasury. The role of WB Treasury in implementation was considered inconsistent with certain provisions of then-current Bank Policy, "Investment Project Financing." A waiver was therefore submitted to OPCS and approved (on April 6, 2021) to allow the WB Treasury to have an expanded implementation role under the Project and intermediate the risk transfer transaction.¹⁶ A second waiver was submitted to the Financial Management Operations Review Committee (FMORC) and approved (on April 21, 2021) to confirm that no interim financial reports or borrower audits would be required given that no project funds would be transferred to the GoJ and the implementation role of the project counterpart, MoFPS. The WB task team also confirmed with the Operations Environmental and Social Review Committee (OESRC) that, given the project design, ESF standards would not apply to proceeds from a potential pay-out of the Cat Bond.

The project has informed the evolution of the Crisis Recovery and Response Toolkit and has contributed to the simplification of similar future projects. Recent changes reflected in the expanded CRR Toolkit (approved by the Board on February 1, 2024) were informed by the Project, in particular, the institutionalization of the option to cover WB DRTT transaction costs via project funds. Thus, it is expected that similar projects in the future would require fewer waivers.¹⁷

B. KEY FACTORS DURING IMPLEMENTATION

Ultimately, no pay-outs were triggered during the risk period of the original Cat Bond. Project implementation consisted of the structuring and placement of the Cat Bond in international capital markets. No named hurricanes breached the centralized air pressure thresholds of the geographic grids during the risk period of the original Cat bond. As of December 2024, no pay-outs had been triggered under the renewed Cat bond (in effect until 2027). The absence

¹⁶ WB Treasury would manage the preparation, structuring and execution of the Cat bond, procuring the external service providers required for the transaction and event calculation agent as well as external counsel and listing agent. The WB was authorized to provide these risk intermediation services under its existing policies applicable to disaster risk management products offered to clients as approved by the Board of Executive Directors, however this represented a novel and expanded role of the WB Treasury in the context of an IPF.

¹⁷ As of December 2024, there has not yet been another RETF IPF approved by the Board to finance a WB-intermediated Cat Bond transaction.



of pay-outs is not unexpected given that the Cat bond was designed to address Jamaica's financial protection gap related to severe tropical cyclone events occurring one in 20 to one in 100 years. As noted above, other instruments in the GoJ's DRF toolkit have paid-out during the risk periods covered by the two Cat bonds, notably a US\$26.6 million pay-out under the GoJ's CCRIF tropical cyclone and excess rainfall insurance policy (which covers a lower layer of risk).

During the risk period of the original Cat Bond, the GoJ decided to renew the Cat bond. The GoJ requested permission to apply the balance of project funds (US\$179,102.57) towards the transaction costs of the Cat bond renewal (as discussed in "Significant Changes During Implementation"). The new Cat bond was issued on May 2, 2024.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

The Project's Overall Rating of Quality of M&E is High. The Project's M&E design, implementation and utilization reflect the novel and straightforward nature of the Project. As summarized above, the Project was not a typical IPF and its implementation consisted of financial transactions intermediated by the WB Treasury to place Cat Bonds in capital markets. The intermediate results indicator (*Cat bond placed in the market: Yes/No*) is a straightforward and mechanistic representation of the key implementation activity of the Project. The original PDO outcome indicator (*Increased insurance coverage from original transaction, amount USD*) adequately reflects the PDO-level outcome of the Project (*to provide increased financial protection against losses arising from tropical cyclones-wind*). Importantly, the indicator and target were designed to be attributable to the grant financing provided under the Project with GriF grant funds (US\$16.365 million), rather than the full donor financing for the Cat bond premiums and transaction costs, which also included US\$5 million from USAID. All Project results indicators fulfill the criteria of good practice results indicators under the SMART framework: Simple, Measurable, Achievable, Relevant, and Time-bound.

Results indicators were appropriately revised during the restructuring to reflect the minor contribution of the balance of Grant funds towards the premiums and transactions costs of the Cat Bond renewal, resulting in the inclusion of a new PDO-outcome indicator (*Increased insurance coverage from renewal transaction, amount USD*) and the upwards revision of the target value for the intermediate results indicator from one to two. The end target-date of the intermediate result indicator (*Cat bond placed in the market*) was changed to June 2024 to include the renewal transaction.

Given that there were no traditional implementation activities under the Project (e.g., procurement, capacity-building, etc.), there were no apparent use cases to utilize the M&E framework to inform ongoing project implementation.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

The Environmental and Social Risk Rating was "Low". Three of ten E&S standards (ESSs) were classified as "Relevant": (i) Assessment and Management of Environmental and Social Risks and Impacts (ESS 1); (ii) Labor and Working Conditions (ESS 2); and (iii) Stakeholder Engagement and Information Disclosure (ESS 10). As of final ISR, ratings of each of these three ESSs was "Satisfactory". The Sexual Exploitation and Abuse/Sexual Harassment (SEA/SH) Risk was "Low".



The Project did not support any physical works or procurement of goods or activities subject to WB procurement rules for IPFs and there were no activities which might have resulted in any adverse impact on human population or the environment. The Project financed risk premium and transaction costs incurred by WB Treasury for the Cat bond only. Procurement conducted by WB Treasury was done according to the Bank's corporate procurement policies. In compliance with the Environmental and Social Commitment Plan (ESCP), the Government of Jamaica (GoJ) finalized and disclosed the Stakeholder Engagement Plan (SEP). The SEP was disclosed by the GoJ on January 24, 2022 and the World Bank on January 26, 2022. Ministry of Finance and the Public Service (MoFPS) designated a Focal Point to monitor the implementation of the ESCP and SEP. Prior to the launch of the Cat bond, the MoFPS shared timely and reliable information to key stakeholders and information continued to be widely available during the project implementation period.

The Grievance Mechanism was implemented by the MoFPS Corporate Communications Branch and the Debt Management Branch (DMB) Investor Relations and Communications unit through its existing external website and social media platforms. Different channels were easily available to potential affected people to submit grievances and queries: Facebook, twitter, email, telephone and in person. The Corporate Communications Manager monitored the GRM and was ready to respond to any queries. However, no grievances were received during the Project. Workers were civil servants of the MoFPS in which there are national Occupational Health and Safety (OHS) protocols and measures in place. The MoFPS Human Resources Department was able to receive complaints from civil servants engaged in the Project through its complaint mechanism.

On Aug 9, 2022, given a low ESRC and satisfactory environmental and social performance, the LCR Regional Standards Advisor released the project from corporate oversight. Given that there are no direct activities by MoFPS, no general Project reports (including on ESCP commitments) were provided to the Bank.

The project's FM risk ratings were Low and ISR ratings were Satisfactory throughout implementation. Project-funded premium and transaction costs for the Cat bonds were directly disbursed to and paid by WB Treasury to the Cat bond investors and service providers. There was no transfer of Project funds to the GoJ. The loan disbursement amount was US\$16,334,361.11. During the Bond term the premium payments from the Trust Fund amounted to US\$16,155,258.54. The excess at maturity of US\$179,102.57 was utilized towards the transaction costs for the new Cat bond issued on May 2, 2024. Given the funds flow arrangements, no interim financial reports or audit reports were required to be submitted by MoFPS to the WB. MoFPS were however required to submit quarterly withdrawal applications for documentation of expenditure in client connection. The agreed submission timeline of 30 days after each calendar quarter end were frequently missed, with noted overdue submission by an average 30-60 days for the majority of quarters. The final documentation of expenditures by the disbursement deadline date of October 21, 2024 was also not met, and was submitted on November 19, 2024. Reassessment of the Disbursement and Financial Information reporting and timelines should be considered for future operations.



C. BANK PERFORMANCE

The Project's Overall Rating of Bank Performance is *Highly Satisfactory*.

The WB ensured quality at entry of the operation. The operation was informed by extensive and robust analytics conducted via a preceding technical assistance project that helped the GoJ to assess (i) the financial impact of disaster events; (ii) existing financial protection gaps; and (iii) the costs and benefits of various disaster risk financing instruments. Analytical support to the GoJ continued during project preparation, including to develop and refine Cat bond prototypes and to assess the market impacts of the Covid-19 pandemic on DRF instrument availability and pricing. Ultimately the WB was responsive to the GoJ's stated priority of pursuing a Cat Bond to diversify their disaster risk finance toolkit. The WB was able to design an innovative project that leveraged and adapted an existing Bank project structure (IPF). The Task Team was also effective in mobilizing funding from donors to fully cover the premium and transaction costs of the original Cat bond.

The WB Treasury supported effective implementation of the operation. During implementation, the WB Treasury managed the preparation, structuring and execution of the Cat bond, procuring the external service providers required for the transaction which included Swiss Re and Aon Securities as lead managers and AIR Worldwide as risk modeling firm and event calculation agent as well as external counsel and listing agent. The GoJ was satisfied with the implementation role of the WB Treasury in structuring and placement of the Cat Bond, which was demonstrated by their continued use of WB Treasury for same services under the Cat Bond renewal and confirmed during discussions held during ICR preparation.

Project supervision was limited in scope given project design. The Project counterparts, MoFPS, had a limited role in implementation, consisting of maintaining monitoring and evaluation arrangements that allow proper reporting and submitting withdrawal requests to the Bank. The task team adequately monitored project performance through the timely completion of periodic Implementation Status and Results Reports (ISRs). The task team also collected all necessary data and reported on a semi-annual basis to the Bank on the use of Grant proceeds and project performance, including information on intermediate project results and progress toward higher-level outcomes. There was minimal turnover in the TTLs and Task Team and management and senior management were engaged throughout Project preparation and implementation.

D. RISK TO DEVELOPMENT OUTCOME

The risk to development outcome is assessed as low. The GoJ's National Natural Disaster Risk Financing Policy was approved by Parliament in June 2023 and remains under implementation through 2026. The GoJ has renewed the original Cat Bond (with largely its own funds) through end-2027 thus continuing to ensure its financial protection against losses arising from severe tropical cyclones-wind and alleviating fiscal constraints and budget volatility arising from natural disasters. The GoJ continues to expand its suite of DRF instruments, for example: (i) in June 2024, the Parliament approved legislation to establish a National Natural Disaster Reserve Fund¹⁸; and (ii) at the time of ICR preparation, a WB Development Policy Loan with a Catastrophe Deferred Drawdown Option (DPF CAT DDO) is under preparation.

¹⁸ An designated contingency fund intended to address disasters that severely affect the economy and lead to a fiscal impact of 1.5 per cent or

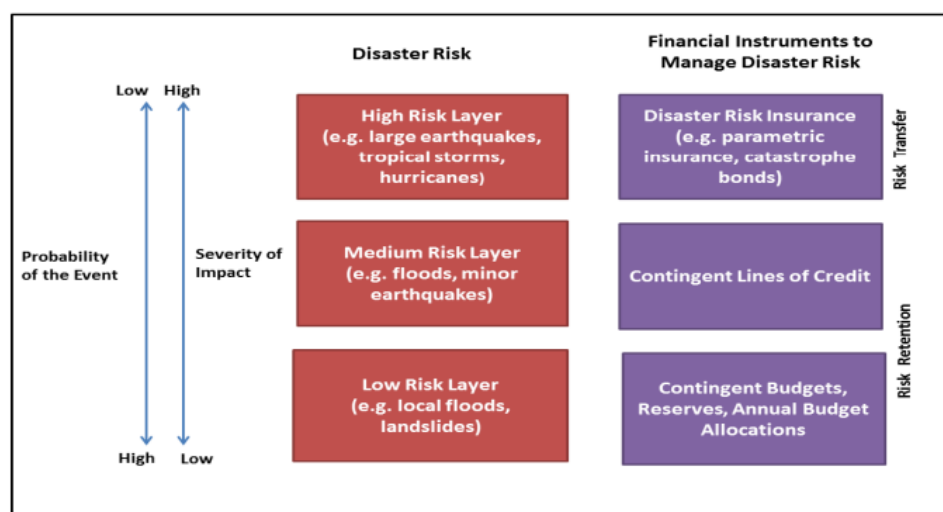


V. LESSONS AND RECOMMENDATIONS

- Disaster risk finance projects should contribute to national DRF strategies underpinned by a risk-layering approach and consisting of a diverse set of complementary DRF instruments.** In the past decade, governments have increasingly prioritized financial protection against natural disasters and developed national DRF strategies that combine a mix of financial instruments based on an assessment of their country's risk profile and associated contingent liabilities and fiscal risks. A comprehensive DRF strategy can secure access to (or "pre-arrange") financing before a natural disaster strikes, ensuring rapid, cost-effective liquidity to finance recovery efforts. The most effective strategies are risk-layered, typically including contingency funds and contingent loans for medium- to high-frequency events of lower severity, complemented by insurance and other types of risk transfer products for major events.

From the outset, the Jamaica Cat Bond project was designed to contribute to Jamaica's National Natural Disaster Risk Financing Policy and risk-layering framework. Indeed, the Cat Bond addressed a specific financial protection gap in Jamaica's DRF strategy, providing protection against low-frequency, high-intensity tropical cyclones-wind events and complementing Jamaica's other pre-arranged DRF instruments including a Contingencies Fund for Natural Disasters, a contingent credit line from IADB, and CCRIF insurance policies. In this sense, the project demonstrates the importance of anchoring World Bank projects in an evaluation of the effectiveness of a country's overall DRF risk-layering approach (considering each country's fiscal, political, and operational context as well as the state of relevant commercial markets) and of encouraging other stakeholders to view a specific DRF intervention in this broader context rather than assessing individual instruments in isolation. This broader view was a success factor in the GoJ's communication efforts with the public regarding the original Cat bonds performance against hurricanes that occurred during the risk period (which did not result in pay-outs).

Figure 3: Jamaica's Multi-Layer Approach to Disaster Risk Financing



Source: Jamaica's National Natural Disaster Risk Financing Policy (2021-2026) Green Paper

In pursuing similar projects in other countries, teams should consider that counterparts may need substantial support for their understanding of risk, including for developing, analyzing, and updating of their risk profile. High-quality,

more of GDP, whether through revenue loss, increased expenditure or both. The Fund complements the National Disaster Fund, which was established under the Disaster Risk Management Act to support mitigation preparedness, rapid early response and recovery.



granular, and up-to-date risk data are essential to developing a country's risk profile and undertaking robust analyses of financial protection gaps, the appropriate mix of DRF instruments, and the specific design of such instruments. The analyses undertaken during the Jamaica Cat bond project preparation to assess the comparative net present value of various DRF instruments available to Jamaica was key to ensuring the project's value-for-money and stakeholder buy-in for the Cat bond instrument. Projects should therefore include substantial investments in data, analytics, and capacity building, while also building flexibility in project design to allow for new information to influence risk financing decisions. Innovative methodologies developed by FCI's global disaster risk finance unit – and informed by the Jamaica Cat bond project – can be usefully applied to other projects, as has been the case in more recent DRF engagements in the Caribbean.¹⁹

- 2. Strong and sustained buy-in among high-level Ministry of Finance counterparts is a key success factor for innovative disaster risk finance projects.** A key success factor throughout the preparation and implementation of the Jamaica Cat Bond project was the high technical capacity of MoFPS and in particular the Minister of Finance (MoF) in understanding and pursuing a risk-layering approach. The MoF was actively involved in all key Project discussions and his strong, sustained, and visible buy-in was instrumental in motivating the Project, achieving consensus among stakeholders, and pushing the Project through the preparation process.

The MoF was also highly effective in explaining the performance of the Cat Bond to Jamaica's population and key stakeholders in the absence of triggered pay-outs following several hurricanes affecting Jamaica during the risk periods. For example, in the aftermath of Hurricane Beryl (under the renewed Cat Bond), the MoF recorded a YouTube video posted to Twitter explaining Jamaica's "multi-layered suite of instruments that can provide financial resources to finance the response and recovery from natural disasters", noting that "it is not designed nor expected that every storm should trigger all instruments" but rather the intent for "some instruments provide resources for each storm". He described the various contingent funds and credits, as well as the risk transfer instruments from CCRIF and the Cat bond, and how they corresponded to the intensity of the recent hurricane. This clear explanation of the Cat Bond's performance relative to a significant storm from the Minister of Finance (underpinned by his sound technical understanding of the instrument and broader portfolio) was important to maintaining support for the instrument and ultimately contributed to its renewal. More generally, clear communication from high-level Ministry of Finance officials are important to ensure that DRF instrument expectations are aligned with reality, helping to ensure that Cat bonds and other parametric DRF instruments can be judged appropriately, and hence can constructively inform future product design, uptake decisions, and implementation.

While the high-capacity of the MoFPS and MoF in Jamaica may be difficult to replicate in other countries, WB task teams should try to identify effective, high-level Ministry of Finance champions for innovative disaster risk finance projects; and conversely consider the implications for project outcomes in countries where such counterparts are not present.

- 3. Transparent design and clear communication are necessary to foster widespread understanding about the role, value and limitations of parametric risk transfer instruments.** Parametric risk transfer instruments may not trigger pay-outs in response to some natural disaster events affecting the country during the risk period. This is particularly true for DRF instruments that cover higher layers of risk (i.e. low frequency events), as in the case of the Jamaica Cat

¹⁹ For example: *Enhancing Financial Resilience to Natural Disasters and Climate Change (P180867)*, *Caribbean insurance solutions for climate and natural disasters (P503360)*, and *Canada Caribbean Resilience Facility - Technical Assistance and Implementation Support for Recovery and Resilience Building Programs (P171256)*.



bond. The absence of pay-outs will not necessarily reflect overly strict triggers or basis risk (i.e., the difference between the index measurement and the loss experience) but rather the deliberate design of the DRF instrument within the broader mix of DRF instruments. However, the *perception* of basis risk, that is, a misunderstanding of the triggers leading to payout expectations, can damage the credibility of the specific instrument and broader risk-layering approach, as well as generate reputational risks for governments and donors.

These misperceptions can be partially mitigated by transparent design of the parametric product, as well as clear, open and proactive communication about the role and expected performance of the product. The relatively simple and transparent Cat-in-the-grid trigger (reflecting physical/observable characteristics of a hurricane) employed for the Jamaica Cat bond was a key element for explaining the performance of the instrument relative to natural disasters affecting Jamaica during the risk periods. Indeed, in the aftermath of Hurricane Beryl, the MoF was able to clearly explain that the Cat Bond would have been triggered if the “center of the storm had passed over the land mass of Jamaica.” There are trade-offs with more complex triggers designs (e.g., modeled loss) but the value of simple and transparent triggers for enabling clear communication with stakeholders and the general population should be appropriately weighted in trigger design considerations. The World Bank – given its analytical capacity, convening role, and intermediation activities – should also play a key role in ensuring that relevant stakeholders (as well as interested onlookers, including in the media) receive clear information about the value and limitations of parametric risk transfer instruments.

4. **Donor-financed disaster risk finance projects can generate impactful “demonstration effects” resulting in greater client capacity to design and finance subsequent DRF instruments.** Jamaica was one of the sixteen countries that benefitted from IBRD’s first ever Cat bond in 2014. However, prior to the project, Jamaica had not independently sponsored a Cat bond. The mobilization of donor funding to finance the premiums and transaction costs of the original Cat bond helped to reduce frictions for the GoJ in entering the Cat bond market for the first time as well as build its institutional capacity for subsequent participation in the market. Indeed, under the Project, Jamaica became the first government in the Caribbean region, and the first of any small island state globally, to independently sponsor a Cat bond. The GoJ’s subsequent renewal of the Cat bond with (almost entirely) its own financial resources suggests that the donor-financed original Cat bond supported under the Project generated an impactful “demonstration effect” with positive medium-term impacts for the GoJ’s management of its DRF portfolio. Donors should be open to supporting a range of DRF instruments, as identified via robust analytics and country-specific policy considerations – such support could span multiple, complementary projects in a given country.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS FRAMEWORK

PDO Indicators by Outcomes

Increased financial coverage against natural disasters								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
Increased insurance coverage from original transaction (Amount(USD))	0.00	Jun/2021	130,000,000.00	Aug/2021	130,000,000.00	Aug/2021	142,000,000.00	Jun/2024
Increased insurance coverage from renewal transaction (Amount(USD))	0	Jan/2024			660,000	Jun/2024	660,000	Jun/2024

Intermediate Indicators by Components

Development and implementation of risk transfer transactions against tropical cyclones-wind effects								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
CAT bond placed in the market (Yes/No)	No	Jun/2021	Yes	Jul/2021	Yes	Jun/2024	Yes	Jun/2024
	Comments on achieving targets		Evidence of CAT bond placed in the market, both in 2021 and 2024					

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. TASK TEAM MEMBERS**

Name	Role
Raquel Alejandra Letelier	Team Leader
Eva M. Gutierrez	Team Leader
Deborah Grace Watson	Financial Management Specialist
Monique Frederica Harper Griffiths	Procurement Specialist
Luciano Gutierrez Wuerzius	Procurement Specialist
Kirk Christopher Nathaniel Brown	Environmental Specialist
Erica Virginia Piber	Social Specialist
Jacqueline Beatriz Veloz Lockward	Counsel
Carla Fabiola Coles	Team Member
Cara Zappala	Team Member
Steen Byskov	Team Member
Jose C. Janeiro	Team Member
Gaukhar M. Kaliaskarova	Team Member
Michael S. Bennett	Team Member
Bart Andre Servaes	Team Member
Fei Wang	Team Member
Daniel Ortiz del Salto	Team Member
Jose Villalobos	Team Member
Anita Tarce	Team Member
Samantha Jane Cook	Team Member
Roberto Dario Bacchini	Team Member
Marlon Rolston Rawlins	Team Member
Naomi Victoria Cooney	Team Member
Sarah K. Kim	Team Member

**B. STAFF TIME & COST**

Stage of Project Cycle	Staff Time & Cost	
	No. of Staff Weeks	US\$ (including travel and consultant costs)
Preparation		
Total	0.00	0.00
Supervision/ICR		
FY22	0.375	2,064.60
FY24	4.266	25,334.16
FY25	2.335	17,552.73
Total	6.98	44,951.49